

SIMATS ENGINEERING
CO2_ASSESSMENT TOOL 1: Analytical Problem Solving

COURSE NAME: Cloud Computing and
Big Data Analytics

COURSE CODE: CSA1522

Name	Jani Dilip
Register Number	192424314

COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)*

Dave's Taxonomy: Manipulation (Level 2)

Total Marks: 50

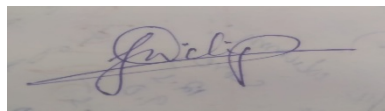
Question Count: 5

Code of Conduct

I **Jani Dilip (192424314)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Assessment Tasks

Analytical Problem Solving

1. Compare Type-1 and Type-2 hypervisors for a cloud datacenter hosting mission-critical applications. Analyze performance, security, and manageability, then justify the better choice.
2. A cloud provider must isolate workloads belonging to finance, healthcare, and student portals on the same hardware. Analyze how virtualization architecture supports isolation and identify two major attack surfaces.
3. Study the following VM host utilization snapshot and recommend corrective actions: CPU 92%, memory 88%, storage I/O latency 35 ms, average VM boot time 170 s. Explain the likely bottlenecks.
4. Analyze how Software Defined Datacenter architecture improves agility compared with a traditional datacenter. Support your answer with compute, storage, and network virtualization considerations.
5. A multi-cloud federation project fails due to identity mismatches between providers. Analyze the issue and propose a secure identity and privacy design using federation concepts.

Analytical Problem Solving Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 – Proficient	Level 2 - Developing	Level 1 - Beginning
Problem Interpretation	20%	Interprets all technical requirements accurately	Interprets most requirements correctly	Partial interpretation	Misinterprets the problem
Analytical Reasoning	25%	Strong reasoning with clear cause-effect analysis	Good reasoning with minor gaps	Basic analysis only	Weak or no analysis
Method Selection	20%	Chooses highly appropriate virtualization and security concepts	Chooses mostly suitable concepts	Partially suitable concepts	Inappropriate approach
Solution Quality	20%	Provides feasible and technically sound solution	Reasonable solution with minor issues	Basic solution with limitations	Unclear or invalid solution
Presentation of Analysis	15%	Well-structured and technically precise	Generally clear	Somewhat unclear	Difficult to follow

Total Marks Awarded:

1. Compare Type-1 and Type-2 Hypervisors for a Cloud Datacenter Hosting Mission-Critical Applications

Introduction

A hypervisor is software that creates and manages Virtual Machines (VMs). It allows multiple operating systems to run on a single physical server. Hypervisors are classified into Type-1 (Bare-Metal) and Type-2 (Hosted) based on how they are installed and operate.

Type-1 Hypervisor

A Type-1 hypervisor runs directly on the physical hardware without requiring a host operating system. Since it has direct access to hardware resources, it provides high performance, better resource utilization, and stronger security. It is widely used in cloud datacentres and enterprise environments.

Key Points

- Runs directly on physical hardware.
- Offers high CPU, memory, and storage performance.

Type-2 Hypervisor

A Type-2 hypervisor runs on top of a host operating system. It depends on the host OS to access hardware resources, which introduces additional overhead and reduces performance. It is mainly used for personal computers, software testing, and development.

Key Points

- Runs on a host operating system.
- Performance is lower due to OS overhead.

Analysis

- Performance: Type-1 hypervisors provide faster processing and better resource utilization because they interact directly with the hardware. Type-2 hypervisors experience additional delays because requests pass through the host operating system.
- Security: Type-1 hypervisors offer stronger security due to their smaller attack surface. Type-2 hypervisors are more vulnerable because any compromise of the host operating system can affect the virtual machines.

Justification

For a cloud datacentre hosting mission-critical applications, the Type-1 hypervisor is the better choice because it provides:

- High performance with minimal overhead.
- Strong security through direct hardware access.

2. Virtualization Architecture for Workload Isolation

Introduction

Virtualization allows multiple virtual machines to run on the same physical server while remaining isolated from one another.

How Virtualization Supports Isolation

- Each workload runs in a separate Virtual Machine (VM).
- Hypervisor isolates CPU, memory, storage, and network resources.
- Virtual networks separate traffic between departments.
- Access control prevents unauthorized communication.
- Snapshots and backup protect workloads independently.

Example

- Finance VM: Stores banking and payment data.
- Healthcare VM: Stores patient records.
- Student Portal VM: Hosts educational applications.

Even though all VMs use the same physical server, they remain isolated.

Major Attack Surfaces

1. Hypervisor Attack
 - Attackers exploit hypervisor vulnerabilities to access multiple VMs.
2. VM Escape Attack
 - Malware escapes from one VM and gains access to the host or other VMs.

3. VM Host Utilization Analysis

Given Data

- CPU Utilization = 92%
- Memory Utilization = 88%
- Storage I/O Latency = 35 ms
- Average VM Boot Time = 170 seconds

Analysis

CPU (92%)

- CPU usage is very high.
- Indicates CPU contention.
- VMs compete for processor time.

Memory (88%)

- Memory is almost full.
- May cause swapping or ballooning.
- Reduces VM performance.

Storage I/O Latency

- Storage latency is high.
- Disk operations become slow.
- Applications experience delays.

VM Boot Time (170 s)

- VM startup is slow.
- Caused by storage bottlenecks and heavy resource utilization.

Likely Bottlenecks

- CPU saturation
- Memory shortage
- Slow storage subsystem
- Resource overcommitment

Recommended Corrective Actions

- Add additional CPU resources or migrate VMs.
- Increase RAM capacity.
- Upgrade storage to SSD
- Balance workloads across hosts.
- Remove unused VMs.
- Enable resource monitoring and auto-scaling.

4. Analyze How Software Defined Datacenter (SDDC) Architecture Improves Agility Compared with a Traditional Datacenter

Introduction

A Software Defined Datacenter (SDDC) is a modern datacentre where compute, storage, and networking resources are virtualized and managed through software. Unlike a traditional datacentre, which relies heavily on physical hardware and manual configuration, an SDDC uses automation to improve flexibility, efficiency, and scalability.

Compute Virtualization

Compute virtualization creates multiple virtual machines on a single physical server, allowing resources to be allocated dynamically based on demand.

Key Points

- Multiple VMs run on one physical server.
- Resources can be allocated or reallocated quickly.
- Supports live migration of virtual machines.
- Improves hardware utilization.
- Reduces the need for additional physical servers.

Storage Virtualization

Storage virtualization combines multiple physical storage devices into a single logical storage pool, making storage management more efficient.

Key Points

- Pools storage resources into a single system.
- Simplifies storage management.
- Allows dynamic storage allocation.
- Improves storage utilization.
- Supports backup and disaster recovery.

Network Virtualization

Network virtualization creates virtual networks that operate independently of physical networking hardware. This enables faster deployment and improved security.

Key Points

- Creates virtual switches and routers.
- Enables network isolation for different applications.
- Allows quick network configuration.
- Improves security through segmentation.
- Supports automated network management.

How SDDC Improves Agility

Because all resources are controlled through software, organizations can respond quickly to changing business needs.

Key Points

- Faster deployment of applications and services.
- Automatic provisioning of resources.
- Rapid scalability according to workload demand.

5. Identity Federation in a Multi-Cloud Environment

Problem

A multi-cloud federation project fails because different cloud providers use different identity systems and authentication methods.

Issues

- Different user identity formats.
- Separate authentication mechanisms.
- Duplicate user accounts.
- Poor access control.
- Increased security risks.

Secure Identity and Privacy Design

Identity Federation

- Establish trust between cloud providers.
- Users authenticate once and access multiple clouds.

Single Sign-On (SSO)

- Users log in once.
- Access multiple cloud services without repeated authentication.

Federation Standards

- SAML (Security Assertion Markup Language) for enterprise authentication.
- OAuth 2.0 for secure authorization.
- OpenID Connect (OIDC) for user authentication.

Privacy Protection

- Encrypt identity data.
- Apply Role-Based Access Control (RBAC).
- Enable Multi-Factor Authentication (MFA).
- Follow the principle of least privilege.
- Maintain audit logs and continuous monitoring.

Benefits

- Secure authentication.
- Better interoperability between providers.
- Improved user experience.